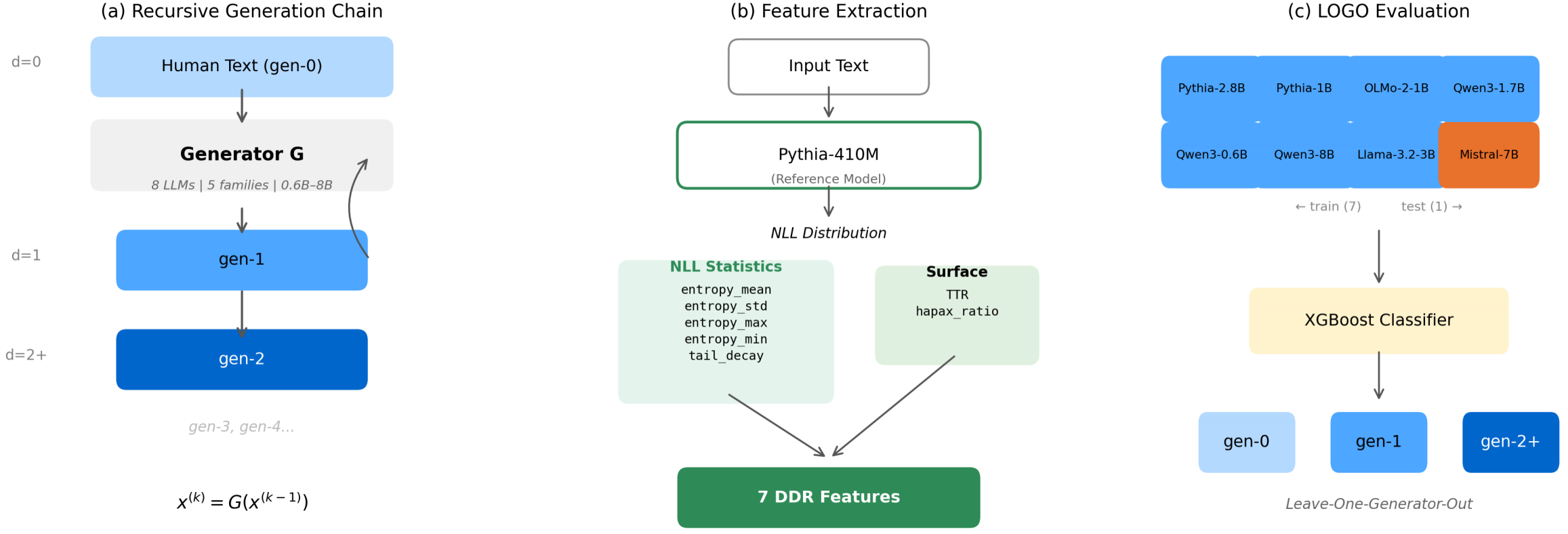

(a) Recursive Generation Chain
d=0
Human Text (gen-0)
Generator G
8 LLMs | 5 families | 0.6B–8B
d=1
gen-1
d=2+
gen-2
gen-3, gen-4...
$x^{(k)} = G(x^{(k-1)})$
(b) Feature Extraction
Input Text
Pythia-410M
(Reference Model)
NLL Distribution
NLL Statistics
entropy_mean
entropy_std
entropy_max
entropy_min
tail_decay
Surface
TTR
hapax_ratio
7 DDR Features
(c) LOGO Evaluation
Pythia-2.8B
Pythia-1B
OLMo-2-1B
Qwen3-1.7B
Qwen3-0.6B
Qwen3-8B
Llama-3.2-3B
Mistral-7B
← train (7)
test (1) →
XGBoost Classifier
gen-0
gen-1
gen-2+
Leave-One-Generator-Out